



A two-herb formula inhibits STAT3 signaling and exerts anti-melanoma effects in cell and animal models

Jun-Kui Li^{a,b,1}, Ji-Yao Chou^{a,b,1}, Cheng-Le Yin^{a,b,1}, Xiu-Qiong Fu^{a,b}, Ying Wu^{a,b},
Ying-Jie Chen^{a,b}, Jing-Xuan Bai^{a,b}, Jia-Ying Wu^{a,b}, Chun Liang^c, Zhi-Ling Yu^{a,b,*}

^a Center for Cancer and Inflammation Research, School of Chinese Medicine, Hong Kong Baptist University, Hong Kong, China

^b Research and Development Centre for Natural Health Products, HKBU Shenzhen Research Institute and Continuing Education, Shenzhen, China

^c Division of Life Science, Center for Cancer Research and State Key Lab of Molecular Neuroscience, The Hong Kong University of Science and Technology, Hong Kong, China

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ABSTRACT

Ethnopharmacological relevance: Malignant melanoma is a fatal cancer. Signal transducer and activator of transcription 3 (STAT3) has been proposed as a therapeutic target of melanoma. An herbal formula *Huai-Hua-San* (HHS) comprising *Sophorae Flos* (SF) and *Gardeniae Fructus* (GF) is traditionally used for treating cancers including melanoma, but the pharmacological basis is unknown.

Aims of this study: This study aimed to investigate the anti-melanoma effects of an ethanolic extract of HHS (HHSE), and explore the involvement of STAT3 signaling in the effects.

Materials and methods: An UPLC-TOF/MS method was developed to control the quality of HHSE. A B16F10 allograft mouse model and three melanoma cell lines (B16F10, A375 and A2058) were used to determine the anti-melanoma effects of HHSE. Dacarbazine (DTIC) and Stattic were used as positive controls. Cell viability was detected using MTT and crystal violet staining assays. Cell apoptosis was analyzed by flow cytometry after the cells were stained with Annexin-V/PI. Cell invasive ability was examined using the transwell assay. Protein levels were determined by Western blotting.

Results: The contents of crocin I, crocin II, quercetin and kaempferol in HHSE were 0.59%, 0.98%, 4.66% and 1.15%, respectively. A clinically relevant dose of HHSE (0.1 g/kg/day, i.g. for 15 consecutive days) significantly suppressed B16F10 tumor growth in mice. HHSE dose-dependently reduced cell viability and dampened invasion of, and induced apoptosis in, melanoma cells. Mechanistic studies revealed that HHSE inhibited the phosphorylation/activation of STAT3 in B16F10 allografts and in cultured melanoma cells. In cell models, HHSE also inhibited the phosphorylation of STAT3 upstream kinases, JAK2 (Tyr1007/1008) and Src (Tyr416), lowered STAT3 nuclear levels, and down-regulated the protein levels of STAT3-targeted molecules. Over-activation of STAT3 in A375 cells significantly attenuated the cytotoxic effects of HHSE.

Conclusions: HHSE exhibits anti-melanoma effects in cell and mouse models. Inhibition of STAT3 signaling contributes to the anti-melanoma mechanisms of HHSE. Our findings lay a groundwork for developing HHSE as a modern agent for melanoma management, and provide pharmacological justifications for the traditional use of HHS in treating melanoma.

1. Introduction

Malignant melanoma is a lethal cancer. Current therapies of melanoma have various disadvantages, such as adverse events, low response rates, and drug resistance [Mackiewicz and Mackiewicz, 2018].

Alternative and/or complementary melanoma therapies are needed.

Constitutive activation/phosphorylation of signal transducer and activator of transcription 3 (STAT3) is a key oncogenic factor of melanoma. Phosphorylation at Tyr-705 is essentially required for STAT3 signaling as it allows STAT3 to dimerize, translocate to the nucleus, and

* Corresponding author. Research and Development Centre for Natural Health Products, HKBU Shenzhen Research Institute and Continuing Education, Shenzhen, China.

E-mail address: zlyu@hkbu.edu.hk (Z.-L. Yu).

¹ These authors contributed equally to this work.

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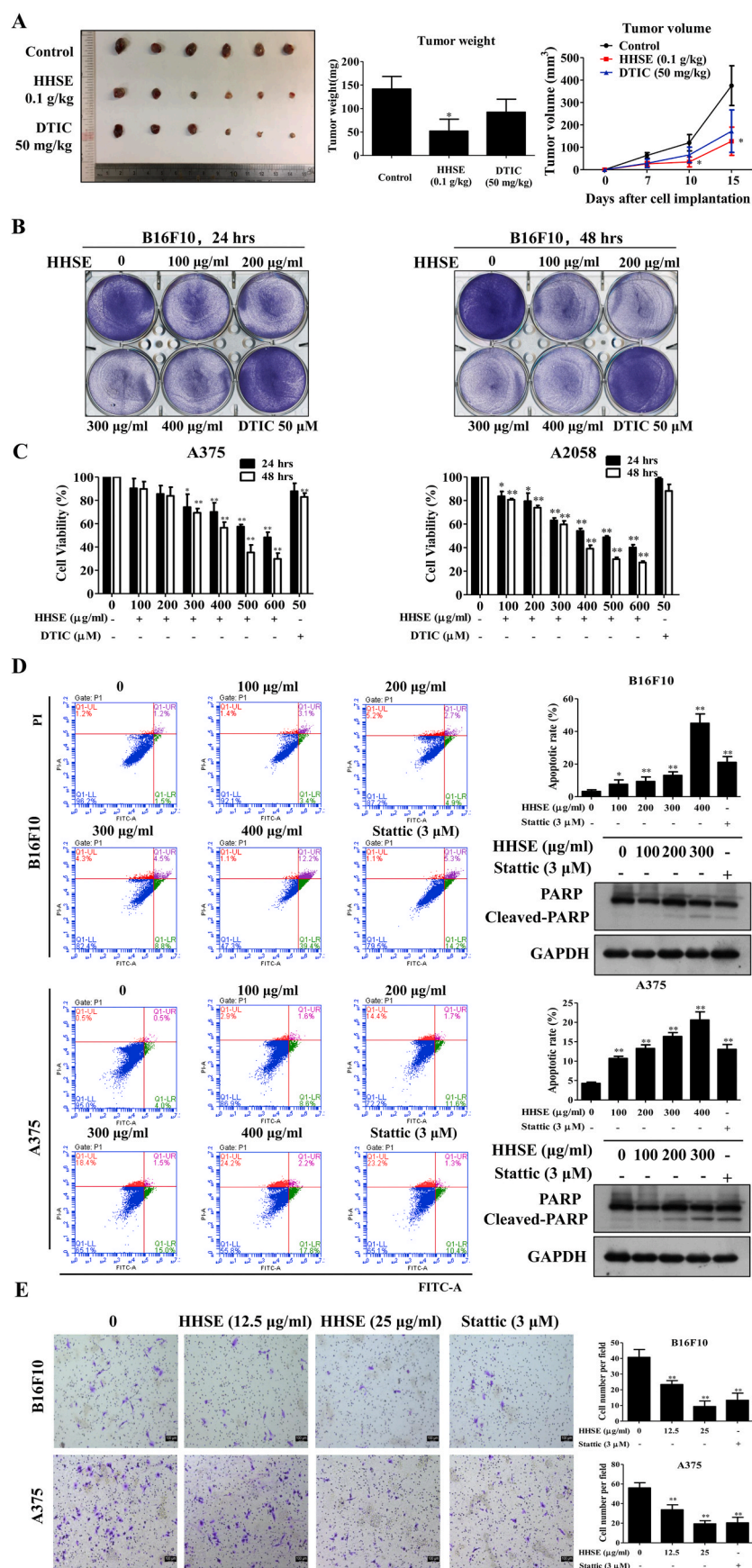


Fig. 1. HHSE exhibits anti-melanoma effects in cell and animal models. (A) HHSE inhibited B16F10 melanoma growth in C57/BL6 mice. Photographs of tumors (left panel), tumor weights (middle panel) and tumor volumes (right panel) are shown. Mice were intragastrically (i.g.) administered with 0.5% CMC-Na (vehicle), HHSE (0.1 g/kg, human equivalent dose) or DTIC (50 mg/kg, positive control) daily for 15 consecutive days. Data are presented as mean $\pm$ SD, $n = 6$. * $P < 0.05$ vs. control group. (B) HHSE reduced the viability of B16F10 cells. B16F10 cells were treated with the indicated concentrations of HHSE or DTIC (50 µM, positive control) for 24 or 48 h. Cell viability was measured by the crystal violet staining assay. (C) HHSE reduced the viability of A375 and A2058 melanoma cells. Cells were treated with various concentrations of HHSE or DTIC (50 µM, positive control) for 24 h or 48 h. Cell viability was measured by the MTT assay. (D) HHSE induced apoptosis in B16F10 and A375 cells. Melanoma cells were treated with the indicated concentrations of HHSE or Stattic (3 µM, positive control) for 48 h. Apoptosis was examined by flow cytometry. Protein levels of PARP were examined by Western blotting. (E) HHSE inhibited the invasive ability of melanoma cells. A375 and B16F10 cells were allowed to invade through a matrigel-coated membrane with 8-mm pores for 24 h in the presence or absence of HHSE (12.5, 25 µg/ml) or Stattic (3 µM, positive control). Representative photos (left panel) and quantification (right panel) of invaded cells are shown. Data are presented as mean $\pm$ SD of three biological repeats, * $P < 0.05$, ** $P < 0.01$ vs. individual control group. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

modulate the transcription of targeted genes [Carpenter and Lo, 2014]. Therefore, agents that can inhibit STAT3 phosphorylation at Tyr-705 site would have anti-melanoma potential.

Huai-Hua-San (HHS), a traditional Chinese medicine (TCM) formula comprising *Sophorae Flos* (SF, the dried flowers and flower buds of *Sophora japonica* L.) and *Gardeniae Fructus* (GF, the dried fruits of *Gardenia jasminoides* Ellis), was documented in a Chinese medicine book *Jing-Yan-Liang-Fang* written 600 years ago. The weight ratio of the two herbs in the formula is 1:1. One of HHS's traditional indications is *Zangdu* that is nowadays diagnosed as cancer. HHS together with other herbs is commonly used for treating melanoma [Chen et al., 2020]. Several compounds, such as quercetin and genipin, occurring in SF and/or GF have been reported to possess anti-melanoma properties and inhibit STAT3 signaling [Cao et al., 2014; Hong et al., 2020]. In this study, we determined whether an ethanolic extract of HHS (HHSE) has anti-melanoma effects and whether inhibiting STAT3 signaling is involved in the effects.

2. Materials and methods

2.1. Preparation of HHSE

SF and GF were provided by the Mr. & Mrs. Chan Hon Yin Chinese Medicine Specialty Clinic and Good Clinical Practice Centre in Hong Kong Baptist University. They were authenticated by Prof. YU Zhiling. Voucher specimens of GF (No. GF20057) and SF (No. SF11060) were deposited at SCM, HKBU. HHS (SF and GF at a weight ratio of 1:1), traditionally taken as herb powder as documented in *Jing-Yan-Liang-Fang*, was extracted with 70% (v/v) ethanol by ultra-sonication to retain both hydrophobic and hydrophilic bioactive components. To enrich bioactive components, the extract was filtered, evaporated and then subjected to a macroporous resin D101 column chromatography which is good in enriching bioactive small molecules in herbal extracts [Ye et al., 2010], eluted with 95% ethanol. The elute was then lyophilized, obtaining HHSE (yield: 6.5%). UPLC-TOF/MS analyses showed that the contents of crocin I, crocin II, quercetin and kaempferol in HHSE were 0.59%, 0.98%, 4.66% and 1.15%, respectively (Supplementary Fig. 1).

2.2. Cell culture

Murine B16F10 and human melanoma A375 and A2058 cell lines were purchased from ATCC (USA). Cells were cultured in Dulbecco's modified Eagle's medium (GIBCO, USA) supplemented with 10% fetal bovine serum (GIBCO, USA) and 1% penicillin/streptomycin at 37 °C in a humidified 5% CO₂ atmosphere [Liu et al., 2019].

2.3. In vivo studies

C57/BL6 mice (Male, 6 weeks old) were obtained from The Chinese University of Hong Kong. All experimental procedures were approved by and conducted in accordance with the guidelines of the Committee on the Use of Human and Animal Subjects in Teaching and Research, Hong Kong Baptist University [No: (17-29) DH/SHS/8/2/6 Pt.]. Immediately after subcutaneous inoculation of B16F10 cells (1×10^6) on the back [Li et al., 2017], mice were randomly allocated into three groups of 6 mice each and intragastrically (i.g.) administered with 0.5% CMC-Na (vehicle), HHSE (0.1 g/kg, human equivalent dose) or DTIC (50 mg/kg, positive control, purchased from Sigma) daily. Considering the 3Rs of the humane use of animals in scientific research, and a long history of clinical application of the formula, we used HHSE's human equivalent dose in the animal experiments. Clinical signs of the mice were monitored daily. Body weights and tumor volume were measured once every 4-7 days. After treatments for 15 days, mice were euthanized with CO₂. Tumors were dissected, weighed and photographed.

2.4. Cell viability and apoptosis assays

B16F10 cells (2×10^5), seeded in 6-well plates, were treated with various concentrations of HHSE for 24 h. B16F10 cells produce melanin, so cytotoxic effects of HHSE on B16F10 cells were examined by crystal violet staining assays. MTT and apoptosis assays were performed as described [Zhu et al., 2019].

2.5. Cell invasion assay

Cell invasion assays were performed using the BD BioCoat™ Matrigel™ invasion chamber (8-μm pore size; Minipore). Melanoma cells were treated with HHSE (12.5, 25 μg/ml) or Stattic (3 μM, positive control) for 24 h. Invaded cells were fixed, stained with 0.5% crystal violet, photographed and counted from 5 randomly selected fields under a microscope (Leica, Germany) [Li et al., 2017].

2.6. Western blot analysis

Proteins in cells and tumor tissues were extracted. Western blotting assay was performed with antibodies against Mcl-1, PARP, SP1, Bcl-xL, MMP-2, MMP-9, phospho-Src (Tyr416), Src, JAK2, phospho-JAK2 (Tyr1007/1008), phospho-STAT3 (Tyr705) and STAT3 (Cell signaling Technology, USA). The amounts of total protein sample loaded were monitored and normalized using an anti-GAPDH antibody (Santa Cruz, CA, USA) [Liu et al., 2019].

2.7. Stable transfection

A375 cells stably expressing the constitutively active STAT3C were established as described previously [Li et al., 2017].

2.8. Statistical analysis

All data are expressed as mean ± SD. Statistical analysis was determined by one-way ANOVA followed by the Dunnett's multiple comparisons using GraphPad Prism version 5.0 software (USA). $P < 0.05$ was regarded as statistically significant.

3. Results and discussion

Results in Fig. 1 showed that treatment with human equivalent dose of HHSE (0.1 g/kg) significantly suppressed B16F10 melanoma growth in mice. The tumor volumes in HHSE-treated group were significantly smaller than those in the control group after dosing for 10 days. The average tumor weight in HHSE-treated group was 45.8% of that in the control group. The positive control DTIC, an approved chemotherapeutic drug for treating malignant melanoma, at human-equivalent dose (50 mg/kg) reduced tumor volume and tumor weight as well.

SF and GF are used as foods in China, suggesting that HHS is safe for human consumption [Chen et al., 2020; He et al., 2016]. HHSE dosing did not significantly affect mouse body weight (Supplementary Fig. 2) and no abnormalities were found in all mice at necropsy, indicating that HHSE has no observable toxicity to melanoma-bearing mice.

Two human melanoma cell lines A375 and A2058 and a murine melanoma cell line B16F10, in which STAT3 are aberrantly activated [Liu et al., 2012], were employed in this study to determine the *in vitro* anti-melanoma effects, and to explore the involvement of STAT3 signaling in the anti-melanoma mechanisms of HHSE. We found that HHSE reduced the viability of each of the three cell types in dose- and time-dependent manners (Fig. 1B and C). The cytotoxic effect of DTIC against melanoma cells was consistent with a previous report [Malvi et al., 2018]. After a 48-hr treatment, HHSE induced apoptosis in B16F10 and A375 cells in a dose-dependent manner, which was validated by the cleavage of PARP (Fig. 1D). Transwell invasion assays showed that B16F10 or A375 cells invaded to the lower membrane of the

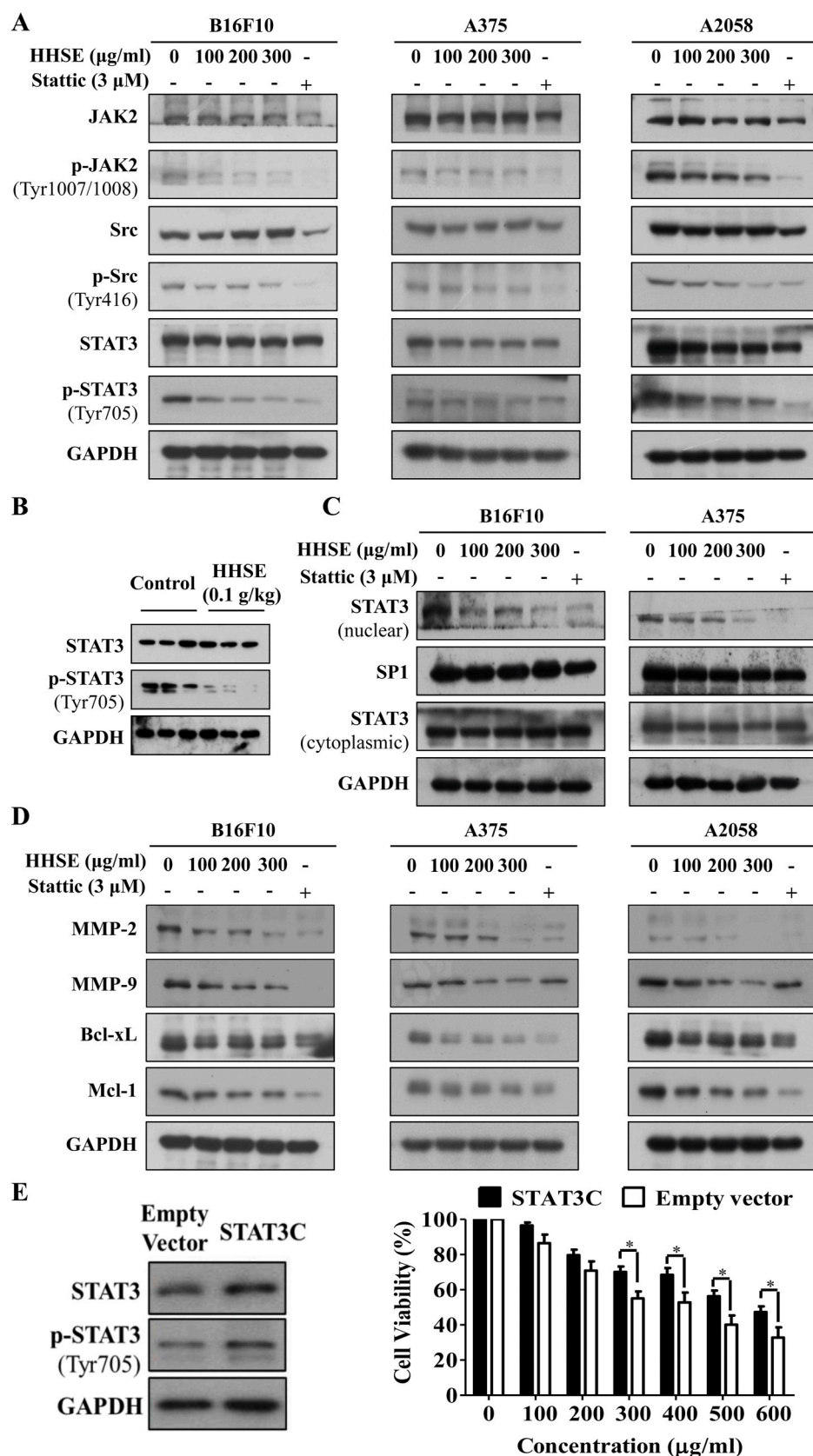


Fig. 2. HHSE suppresses STAT3 signaling in melanoma cell and mouse models. (A) HHSE inhibited STAT3 signaling in melanoma cells. (B) HHSE inhibited STAT3 phosphorylation in B16F10 tumors. (C) HHSE lowered the STAT3 nuclear protein levels in melanoma cells. (D) HHSE downregulated the protein levels of STAT3-targeted molecules in melanoma cells. In (A), (C) and (D), melanoma cells were treated with the indicated concentrations of HHSE or Stattic (3 μM, positive control) for 48 h. (E) Overexpression of STAT3C diminished the cytotoxic effects of HHSE in A375 cells. A375 cells stably expressing STAT3C (a constitutively active STAT3 variant) or the empty vector were treated with the indicated concentrations of HHSE for 48 h. Cell viability was determined by the MTT assay. The left panel shows the protein levels of STAT3 and p-STAT3 in the stable cell lines. The right panel shows the cell viability assay results. In (A), (C)–(E) data are from three biological repeats; in (B), $n = 3$; * $P < 0.05$.

chamber were significantly and dose-dependently fewer in HHSE-treated groups than in the control group (Fig. 1E), suggesting that HHSE inhibited the invasive capability of melanoma cells. Stattic, a STAT3 inhibitor, has been reported to inhibit melanoma cell survival and invasion [Fu et al., 2018; Liu et al., 2019]. It was used as a positive control in cell apoptosis and invasion assays. As found in previous studies [Fu et al., 2018; Liu et al., 2019], Stattic significantly triggered apoptosis and mitigated invasion of B16F10 and A375 cells. These findings indicate that HHSE has anti-melanoma effects in cell and mouse models, and has no overt toxicity in mice.

Inhibiting STAT3 activation has been reported to suppress melanoma growth [Cao et al., 2014]. Here, we found that HHSE lowered the protein levels of phospho-STAT3 (Tyr705) in melanoma cells (Fig. 2A) and B16F10 allograft tumors (Fig. 2B). STAT3 can be activated by its upstream kinases, such as JAK2 and Src [Lesinski, 2013]. We observed that HHSE downregulated protein levels of phospho-Src (Tyr416) and phospho-JAK2 (Try1007/1008) in melanoma cells (Fig. 2A). As a transcriptional factor, activated STAT3 gains entrance to the nucleus to transcribe its targeted genes, such as cell survival related genes Bcl-xL and Mcl-1, and cell invasion related genes MMP-2 and MMP-9 [Lee et al., 2019]. Immunoblotting results showed that HHSE lowered the nuclear STAT3 protein levels (Fig. 2C), and downregulated the protein levels of Mcl-1, Bcl-xL, and MMP-2/9 in melanoma cells (Fig. 2D). Similar results were observed in Stattic-treated cells (Fig. 2A, C and 2D). To determine whether inhibiting STAT3 contributes to the anti-melanoma effects of HHSE, A375 cells persistently expressing over-activated STAT3 (STAT3C) were used. Results showed that over-activation of STAT3 in A375 cells significantly attenuated the cytotoxic effects of HHSE (Fig. 2E). Apart from melanoma, compounds derived from HHSE are also effective in suppressing other cancers, such as colorectal and liver cancer [Zhang et al., 2017]. STAT3 is also aberrantly activated in these cancers. Whether HHSE has a potential to treat these cancers is to be investigated.

4. Conclusions

The present study demonstrated that HHSE exhibits anti-melanoma effects in animal and cell models, and inhibiting STAT3 signaling is involved in the effects. These findings lay a groundwork for developing HHSE as a new modern anti-melanoma agent, and provide pharmacological justifications for the traditional use of HHS in treating melanoma.

Chemical compounds studied in this article

Crocin I (PubChem ID: 5281233), Crocin II (PubChem ID: 5281233), Quercetin (PubChem ID: 5280343) and Kaempferol (PubChem ID: 5280863).

Authors' contributions

JKL, JYC and CLY conducted experiments, analyzed data and drafted the manuscript. XQF, YW, YJC, JXB and JYW participated in experiments; CL provided suggestions and edited the manuscript; ZLY conceived and supervised this study.

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Declaration of competing interest

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2020.113671>.

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